



Skills

Mechanical: Siemens NX, SolidWorks (Surface Modelling, Assemblies), GD&T, DFM/DFA, Fixtures & Jigs, CAD Drawings & BOMs, 3D Printing, Mill
Electrical & Hardware: Altium Designer, Schematic Design, HIL Test Benches, Soldering, Hot Air Rework, Oscilloscope, ESP32, CAN Bus
Programming: Python, C++, Embedded C, ROS/DDS, MuJoCo, Sim-to-Real RL, Linux, FreeRTOS, MicroPython, Shell/Bash, Git
Software & Tools: Linux, Git, GitLab CI/CD, VS Code, Putty, SolidWorks Simulation, MATLAB, Microsoft Office Suite, Google BigQuery



Work Experience

Geotab | Embedded Developer Intern

May 2026 – Aug 2026

Fleet Management & Telematics

Oakville, CA

- Integrated 29-DOF Unitree G1 humanoid robot with Inspire dexterous hands for automated warehouse material handling.
- Developed Python GUI with individual joint slider controls and real-time trajectory recorder to automate motion routines.
- Implemented live camera overlay using Google MediaPipe to track hand landmarks and mimic finger postures on Inspire hands.
- Engineered Siemens NX CAD mounts for RealSense depth cameras, touch display, and integrated hardware safety E-Stop circuitry.
- Trained and deployed sim-to-real PPO reinforcement learning policy in MuJoCo for 7-joint arm trajectory tracking over DDS.

Geotab | System Verification Intern (HIL Team)

Aug 2025 – Dec 2025

Fleet Management & Telematics

Oakville, CA

- Developed automated Hardware-in-the-Loop (HIL) test fixtures to validate GNSS, sleep states, and CAN bus vehicle telemetry.
- Integrated prototype telematics units into daily automated HIL test pipelines, automating regression execution via Python scripts.
- Improved automated test suite coverage by 80%, providing critical diagnostic data to expedite embedded firmware bug fixes.

Geotab | Embedded Developer & QA Intern

Jan 2025 – Apr 2025

Fleet Management & Telematics

Oakville, CA

- Assembled test fixtures and programming jigs on production line; performed end-of-line QA validation and flashed device firmware.
- Troubleshooted electromechanical issues on returned and prototype devices with cross-functional firmware, EE, and mech teams.
- Developed Python and GUI tools to automate device logging and test uploads, improving efficiency for cross-functional teams

WatArrow & SAE Aero Design East 2025 | Electromechanical Member

Sept 2024 – Apr 2025

Waterloo Student Design Team & International Competition

Waterloo, CA

- Prototyped foam control surfaces, manufactured modular balsa wings, and performed carbon fiber layup for custom landing gear
- Wired servo motors to flight computer using modular crimped connections and created detailed electrical diagrams.
- Competed at SAE Aero East 2025; fixed flight computer issue causing flap creep, resulting in 3rd place in mission points.



Engineering Projects

7-DOF QDD Robotic Arm | Personal Project

Jan 2026 – Present

- Engineered a 7-DOF quasi-direct-drive (QDD) robotic arm in SolidWorks achieving a 1:1 payload-to-weight ratio (10 kg payload).
- Applied surface modelling for link housings; designed modular actuators and interlocking spigot joints with heat-set inserts.
- Complex CAD assembly with multiple subassemblies, configurations and sub-configurations (375+ parts).
- Implemented distributed CAN bus architecture with custom ESP32 nodes to execute real-time closed-loop motor trajectory control.

10-DOF Robotic Hand | Personal Project

Dec 2025 – Present

- Modeled a 10-DOF robotic hand in SolidWorks, validating grasping kinematics via dynamic mates and adjusting for tolerance.
- Routed high-strength tendon cables through low-friction conduits with spring returns for compliant, multi-object grasping.
- Applied advanced surface modelling (lofts & boundary surfaces) to sculpt an ergonomic palm housing 10 micro-servos.
- Engineered a 10 A transient-safe power rail with 4700 μ F capacitance and offloaded 50 Hz PWM generation to PCA9685 via I2C.

Custom PCB Fixture | Internship Project

Mar 2025 – Apr 2025

- Developed modular R&D test fixtures and programming jigs in SolidWorks to validate prototype PCBs in laboratory setups.
- Modeled 20+ custom CAD parts, selected COTS toggle clamps and pogo pins, and generated detailed drawings and complete BOMs.
- Prototyped 3D printed alignment guides and spring-loaded pogo pin test beds; validated tolerances and cut fixture cost by 95%.

Automatic Door Locker | Personal Project

Sept 2025 – Dec 2025

- Designed a universal door actuator in SolidWorks, simulating gear reduction and tolerancing to maximize dynamic torque output.
- Drafted schematics and routed 2-layer PCB in Altium Designer integrating an LM2596S regulator, INA219 monitor, and TMC2209 driver.
- Built MicroPython web server on RP2040 Pico with sensorless stall detection for auto-calibrating zero-position end stops.

Airfoil Wind Tunnel | WatArrow Design Team Project

Oct 2024 – Dec 2024

- Designed wind tunnel test rig in SolidWorks with custom structural mounts for load cell and stepper motor instrumentation.
- Fabricated test rig with machined aluminum framing, 3D printed nozzles, and COTS sensors; conducted aerodynamic testing.



Education

University of Waterloo

Sep 2024 – Present

Bachelor of Engineering in Mechatronics Engineering

Waterloo, CA

- Awarded University of Waterloo President's Scholarship of Distinction & Academic Standing: Excellent Standing
- **Relevant Coursework:** Real-Time Systems, Sensors & Instrumentation, Linear Systems & Signals, Numerical Methods, Advanced Calculus